Name: _____

TOPIC TEST 2

Chapter 3 Applications of derivatives

- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Five multiple-choice questions (5 marks).
- Part B: Six short-answer questions (39 marks).
- Part C: Two analysis questions (16 marks).

Part A

Multiple-choice questions

Five multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Find the value of $27^{\frac{2}{3}}$ and with the use of derivatives find an approximation for $29^{\frac{2}{3}}$.

A 9.2

B 9.33

C 9.66

D 9.22

E 9.44

[1 mark]

2 Given that $f(x) = -2x^3 - 3x^2 + 4x - 1$, the value of $f''(-2)$ is:

A -30

B -8

C -5

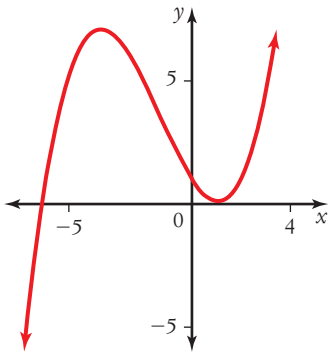
D 18

E 30

[1 mark]

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3 For the graph shown, which is true at $x = -5$?



A $\frac{d^2y}{dx^2} > 0$, $\frac{dy}{dx} > 0$

B $\frac{d^2y}{dx^2} > 0$, $\frac{dy}{dx} < 0$

C $\frac{d^2y}{dx^2} < 0$, $\frac{dy}{dx} > 0$

D $\frac{d^2y}{dx^2} < 0$, $\frac{dy}{dx} < 0$

E $\frac{d^2y}{dx^2} = 0$, $\frac{dy}{dx} > 0$

[1 mark]

4 Find two non-negative numbers whose sum is 24 so that the product of one number and the square of the other number is a maximum.

A 12 and 12

B 8 and 16

C 4 and 20

D 3 and 21

E 9 and 1

[1 mark]

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5 A closed box is to be constructed with a rectangular base with its length 3 times its width. If the volume of the box is 36 m^3, determine the dimensions that will minimise the cost to build the box.

A 6 m long by 2 m wide by 3 m high

B 9 m long by 3 m wide by 1⅓ m high

C 3 m long by 1 m wide by 12 m high

D 4.5 m long by 1.5 m wide by 5⅓ m high

E 7.5 m long by 2.5 m high by 1.92 m high

[1 mark]

Part B

Short-answer questions

Six short-answer questions
34 marks
Show your working where appropriate.

1 Find the first and second derivatives of $f(x) = \dfrac{2x - 7}{x + 3}$.

[4 marks]

2 Find the value of n if $f(x) = 2nx^3 - 5x^2$ and $f''(2) = 2$.

[4 marks]

3 Find any stationary points on the curve $y = x^3 - 4x^2 + 4x + 1$ and determine their nature. Sketch the curve.

[10 marks]

4 Find the dimensions of the rectangular building with the largest area that can be placed on a right-triangular building lot, facing one of the perpendicular sides, if the perpendicular sides are 80 m and 150 m in length.

[7 marks]

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5 A man is on a dock on the west shore of a large lake. He is 12 km from the opposite shore and he needs to reach a point on the opposite shore 20 km south. He can hire a motorboat for $3.00 per hour, for the time it is actually used. The motor boat can travel at 30 km/h. When he reaches the other shore he can catch a bus which costs $0.06 /km. What is the least amount he must pay for the trip?

[6 marks]

6 The volume of a cone is $\frac{8\pi}{3}$ m^3. Find the minimum length of the slant edge?

[8 marks]

Part C

Analysis questions

Two analysis questions
16 marks
Show your working where appropriate.

1 The curve $y = ax^4 + bx^3 - 24x^2 + 11x - 9$ has points of inflection at $x = -2$ and $x = 1$. Find the values of a and b.

[7 marks]

2 A store has been selling 200 Xboxes a week at $390 each. A market survey was undertaken. The results indicated that for each $10 rebate offered to buyers, the number of units sold will increase by 20 a week. How large a rebate should the store offer to maximise its profit?

[9 marks]

[Total marks: 60]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 $f(x) = x^{\frac{2}{3}}$

$f'(x) = \frac{2}{3}x^{-\frac{1}{3}}$

When $x = 27, f(27) = 27^{\frac{2}{3}} = 9$

$f'(27) = \frac{2}{3} \times 27^{-\frac{1}{3}}$

$= \frac{2}{3} \times \frac{1}{3}$

$= \frac{2}{9}$

$\delta y \quad \frac{dy}{dx} \times \delta x$

When $x = 29, \delta x = 2$

$\therefore$ **E**

$\delta y \quad \frac{2}{9} \times 2$

$= \frac{4}{9}$

$29^{\frac{2}{3}} \quad (27 + 2)^{\frac{2}{3}}$

$= 9 + \frac{4}{9}$

9.44

2 $f(x) = -2x^3 - 3x^2 + 4x - 1$

$f'(x) = -6x^2 - 6x + 4$

$f''(x) = -12x - 6$

$f''(-2) = -12 \times (-2) - 6$

$= 24 - 6$

$= 18$

$\therefore$ **D**

3 Curve is concave downwards: $\frac{d^2y}{dx^2} < 0$

Curve is increasing: $\frac{dy}{dx} > 0$

$\therefore$ **C**

4 Let the two numbers be x and $24 - x$.

$f(x) = (24 - x)\,x^2$

$= 24x^2 - x^3$

$f'(x) = 48x - 3x^2$

$= 0$ when $48x - 3x^2 = 0$

$3x\,(16 - x) = 0$

$x = 0, 16$

$f''(x) = 48 - 6x$

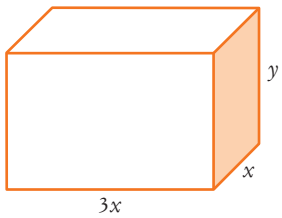
When $x = 16, f''(16) = 48 - 6 \times 16 < 0$

maximum when $x = 16$

Numbers are 16 and 8

$\therefore$ **B**

5



Volume $= 3x^2y$

$3x^2y = 36$

$y = \dfrac{12}{x^2}$

Surface area $= 6x^2 + 8xy$

$A = 6x^2 + 8x \times \dfrac{12}{x^2}$

$= 6x^2 + \dfrac{96}{x}$

$\dfrac{dA}{dx} = 12x - \dfrac{96}{x^2}$

$= 0 \quad$ when $12x = \dfrac{96}{x^2}$

$x^3 = 8$

$x = 2$

$\dfrac{d^2A}{dx^2} = 12 + \dfrac{192}{x^3} < 0 \quad$ for all $x > 0$

$\therefore$ minimum when $x = 2$

width $= 2$ m, length $= 6$ m, height $= 3$ m

$\therefore$ **A**

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Part B – Short-answer questions

1 $f(x) = \dfrac{2x - 7}{x + 3}$

$f'(x) = \dfrac{(x + 3) \times 2 - (2x - 7) \times 1}{(x + 3)^2}$ **[1 mark]**

$= \dfrac{2x + 6 - 2x + 7}{(x + 3)^2}$

$= \dfrac{13}{(x + 3)^2}$ **[1 mark]**

$= 13(x + 3)^{-2}$

$f''(x) = -2 \times 13(x+3)^{-3}$ **[1 mark]**

$= \dfrac{-26}{(x + 3)^3}$ **[1 mark]**

2 $f(x) = 2nx^3 - 5x^2$

$f'(x) = 6nx^2 - 10x$ **[1 mark]**

$f''(x) = 12nx - 10$ **[1 mark]**

$f''(2) = 2$

$12n \times 2 - 10 = 2$ **[1 mark]**

$24n = 12$

$n = 0.5$ **[1 mark]**

3 Maximum at $\left(\dfrac{2}{3}, 2\dfrac{5}{27}\right)$ and minimum at (2, 1)

$y = x^3 - 4x^2 + 4x + 1$

$y' = 3x^2 - 8x + 4$ **[1 mark]**

$y' = 0$ when $3x^2 - 8x + 4 = 0$

$(3x - 2)(x - 2) = 0$

$x = \dfrac{2}{3}, 2$ **[1 mark]**

$y'' = 6x - 8$ **[1 mark]**

When $x = \dfrac{2}{3}$

$y'' = 6 \times \dfrac{2}{3} - 8 < 0$ **[1 mark]**

maximum at $\left(\dfrac{2}{3}, 2\dfrac{5}{27}\right)$ **[1 mark]**

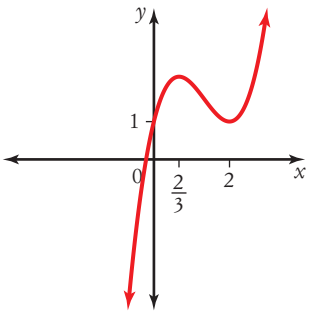
When $x = 2$

$y'' = 6 \times 2 - 8 > 0$ **[1 mark]**

minimum at (2, 1) **[1 mark]**

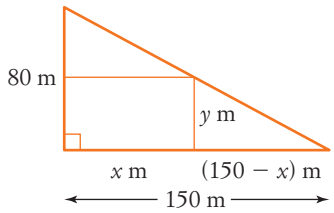
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When $x = 0, y = 1$



1 mark for maximum
1 mark for minimum, 1 mark for y-intercept

4



Using similar triangles:

$$\frac{150 - x}{y} = \frac{150}{80}$$ **[1 mark]**

$$y = \frac{80(150 - x)}{150}$$

$$= 80 - \frac{8x}{15}$$ **[1 mark]**

$$\text{Area rectangle} = \left(80 - \frac{8x}{15}\right) \times x$$

$$A = 80x - \frac{8x^2}{15}$$ **[1 mark]**

$$\frac{dA}{dx} = 80 - \frac{16x}{15}$$ **[1 mark]**

$$= 0 \quad \text{when} \quad 80 = \frac{16x}{15}$$

$$x = \frac{80 \times 15}{16}$$

$$x = 75$$ **[1 mark]**

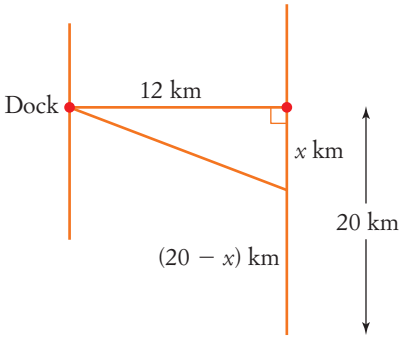
$$\frac{d^2A}{dx^2} = -\frac{16}{15} < 0$$

maximum at $x = 75$ **[1 mark]**

when $x = 75$, $y = 80 - \frac{8 \times 75}{15} = 40$ **[1 mark]**

The building is 75 m long by 40 m wide.

5 $2.16



Distance travelled by boat $= \sqrt{144 + x^2}$, distance travelled by bus $= 20 - x$ **[1 mark]**

$$\text{Cost} = 3 \times \frac{\sqrt{144 + x^2}}{30} + (20 - x) \times 0.06$$ **[1 mark]**

$$C = \frac{\sqrt{144 + x^2}}{10} + 1.2 - 0.06x$$

$$C' = \frac{x}{10\sqrt{144 + x^2}} - 0.06$$ **[1 mark]**

$$= 0$$

when $\dfrac{x}{10\sqrt{144 + x^2}} = 0.06$ **[1 mark]**

$$x = 0.6\sqrt{144 + x^2}$$

$$x^2 = 0.36\left(144 + x^2\right)$$

$$0.64x^2 = 0.36 \times 144$$

$$x^2 = \frac{0.36 \times 144}{0.64}$$

$$= 81$$

$$x = 9$$ **[1 mark]**

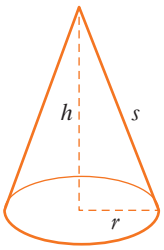
$$\text{Cost} = \frac{\sqrt{144 + 9^2}}{10} + 1.2 - 0.06 \times 9$$

$$= \$2.16$$ **[1 mark]**

6 Volume $= \frac{1}{3}\pi r^2 h$

$24\pi = \frac{1}{3}\pi r^2 h$ **[1 mark]**

$r^2 = \frac{72}{h}$ **[1 mark]**



$s^2 = r^2 + h^2$

$s^2 = \frac{72}{h} + h^2$ **[1 mark]**

Minimum slant when s^2 a minimum.

$\frac{ds^2}{dh} = -\frac{72}{h^2} + 2h$ **[1 mark]**

$= 0$

when $2h = \frac{72}{h^2}$ **[1 mark]**

$h^3 = 36$

$h = \sqrt[3]{36}$ **[1 mark]**

$s^2 = \frac{72}{h} + h^2$ **[1 mark]**

$s^2 = \frac{72}{\sqrt[3]{36}} + \left(\sqrt[3]{36}\right)^2$

$s \quad 5.7$

The minimum length of the slant edge is about 5.7 m.

Part C – Analysis questions

1 $y = ax^4 + bx^3 - 24x^2 + 11x - 9$

$y' = 4ax^3 + 3bx^2 - 48x + 11$ **[1 mark]**

$y'' = 12ax^2 + 6bx^2 - 48$ **[1 mark]**

$= 0$

When $12ax^2 + 6bx^2 - 48 = 0$ **[1 mark]**

$2ax^2 + bx^2 - 8 = 0$

When $x = -2$

$8a + 2b = 8$

$4a + b = 4 \quad \ldots\ldots (1)$ **[1 mark]**

When $x = 1$

$2a + b = 8 \quad \ldots\ldots (2)$ **[1 mark]**

Solving (1) and (2)

$2a = -4$

$a = -2$ **[1 mark]**

$b = 12$ **[1 mark]**

2 Let the number of Xboxes sold be n.

The weekly increase in sales is $n - 200$. **[1 mark]**

For each increase of 20 Xboxes sold, the price decreases by \$10, i.e. each Xbox reduces the price by \$0.50.

The price of each unit is: $P = 390 - 0.5(n - 200)$ **[1 mark]**

$= 490 - 0.5n$

The income from the sale of n units is: $I = n \times (490 - 0.5n)$ **[1 mark]**

$= 490n - 0.5n^2$

$\frac{dI}{dn} = 490 - n$ **[1 mark]**

$= 0 \quad$ when $n = 490$ **[1 mark]**

$\frac{d^2I}{dn^2} = -1 < 0$ **[1 mark]**

$\therefore$ maximum when $n = 490$

$P = 490 - 0.5n$

$P = 490 - 0.5 \times 490$ **[1 mark]**

$= 245$ **[1 mark]**

Rebate $= \$390 - \$\,245 = \$145$ **[1 mark]**